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Confidence-Aware Prediction and Portfolio Construction for Cryptocurrency Pump-and-Dump Events

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ABSTRACT Predicting the target of a scheduled cryptocurrency pump-and-dump event requires distinguishing one manipulated asset from hundreds of candidates. Correct rank alone is insufficient when model scores also determine asset-selection probabilities or portfolio weights: a narrowly top-ranked target can still receive little probability or capital. We compare equal and score-proportional allocation on the same selected assets and use this portfolio perspective to motivate event-level QuerySoftMax training, which rewards both correct ordering and separation of the true target's logit from competing logits. The study uses 517 Binance BTC-pair events, 81 pre-decision features, and chronological training, tuning, comparison, and test partitions of 310, 52, 103, and 52 events. We evaluate target inclusion under weighted sampling without replacement by deterministic numerical integration, eliminating portfolio-resampling noise from the selection metric. QuerySoftMax attains the highest comparison inclusion AUC, 0.4301, and a test AUC of 0.5036. Paired whole-event bootstrap tests show superiority over six alternatives on both partitions after Holm correction ($p_{\mathrm{Holm}} = 0.0028$); its difference from SMOTE is inconclusive, and SMOTE has the higher test AUC. QuerySoftMax also has the highest median return among the eight sampled, score-proportionally allocated five-asset portfolios on both partitions. These portfolio simulations are descriptive rather than significance tests. The findings support confidence-sensitive objectives for this allocation rule, subject to the retrospective sample design and execution assumptions.

INDEX TERMS Confidence-sensitive learning, Learning to rank, Market manipulation, Market microstructure, Portfolio allocation, Weighted sampling

I. INTRODUCTION

A. WHAT ARE PUMPS AND DUMPS

PUMP-AND-DUMP schemes (P&Ds) are a type of fraudulent activity that involves artificially inflating the price of a cryptocurrency through coordinated buying efforts, followed by a rapid selling to unsuspecting investors. This phenomenon is often characterized by an abrupt and significant increase in trading volume, followed by a subsequent decline in price as the original buyers liquidate their positions.

These schemes typically involve experienced traders and market manipulators who orchestrate the operation through various channels, including online forums and social media platforms such as Telegram channels. The manipulation is designed to create a false sense of urgency and scarcity, which attracts inexperienced investors who may be unaware of the true nature of the event.

A typical pump-and-dump scenario can be observed in Fig. 1, which shows a clear example of this phenomenon. The figure illustrates price movements associated with a specific token VIB traded against BTC on Binance exchange, highlighting the abrupt spike in trading activity following the announcement of the token's release. This sudden increase in volume is often followed by a rapid decline in price as insiders sell their positions to new buyers.

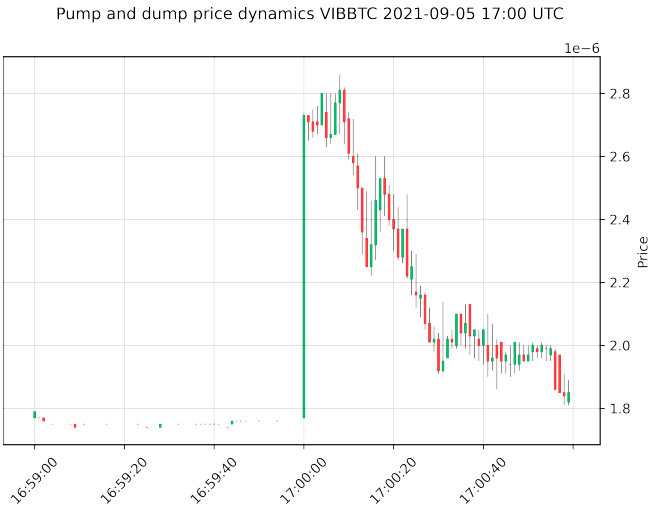


FIGURE 1. **VIBBTC pump and dump manipulation carried out on Binance**

B. RELATED WORK

The phenomenon of pump-and-dump schemes has been extensively studied in the context of cryptocurrencies, with research efforts spanning multiple approaches and methodologies.

One of the pioneering studies on this topic was [9], which introduced a comprehensive framework for understanding pump-and-dump operations. The authors highlighted the targeting of small market cap cryptocurrencies due to their lower trading volumes, utilizing data from 5 prominent crypto exchanges: Binance, Bittrex, Kraken, Kucoin, and Lbank. Their analysis covered pumps history of 1.5 years, with price and volume data being used as the primary sources. Subsequent research efforts have focused on leveraging machine learning (ML) algorithms to enhance P&D detection accuracy. La Morgia et al.'s work [10], [11] employed random forest algorithms, achieving improved results compared to traditional rule-based approaches. The authors in [4] further increased the effectiveness of La Morgia's work by incorporating Deep Neural Networks (LSTMs and Transformers) as anomaly detection algorithms. More recently, in [6] the authors built on these findings by studying a similar pump dataset between 2021 and 2022. They developed an innovative approach that employed resampling methods, such as SMOTE, combined with price and volume features and random forest classification to detect pump events 1 hour in advance.

In addition to pump detection, other research has focused on predicting the maximum price move during known pump events. Nghiem et al. [12] used market data and social sentiment with deep neural networks (LSTMs and CNNs) and achieved promising results. However, their analysis highlighted a potential limitation of using sentiment signals: historical bias may be introduced due to the lack of precise snapshots of historical social data.

A relevant example of work addressing the problem of predicting price manipulation targets is the study [16]. This research proposed an approach using random forest algorithms to identify potential pump and dump schemes among listed tokens. The authors analyzed a dataset of 180 pump events from the Cryptopia exchange, incorporating features such as price and volume data, global market capitalization data from Coinmarketcap, and token ratings by exchange. To enhance model robustness, the study employed filtering techniques to exclude pumps with suspicious announcement times that deviated significantly from full hours. Moreover, the authors validated their classification model through backtesting a portfolio strategy based on predicted pump targets. However, it is worth noting that the employed trading strategy relied on unrealistic assumptions, such as guaranteed profits of half the maximum price move during the pump and zero losses for false positive predictions.

A related study was conducted in [8], where authors enhanced their approach by incorporating channel embeddings into their model. They also utilized standard price and volume metrics, as well as external data sources such as market capitalization and social media information (e.g., Twitter followers, Reddit subscribers, Alexa rank). The analysis was performed on a dataset comprising 445 pump announcement events. However, due to the presence of duplicate pump events across multiple channels, the actual number of unique pump events studied is smaller than expected. Furthermore, the use of social data from different sources raises concerns about historical bias, as these metrics may have changed after the fact, potentially affecting the accuracy of the results.

Regarding market manipulation in a broad sense, several papers have also been published recently, for example [5], which studied the manipulation of trading volumes, also known as wash trading.

With the increasing popularity of ML applications in finance, in recent years we have seen many articles on predictions using the market microstructure, for example [2] and [13]. Promising results have also been published with applications of ranking models to finance [14]. Taking into account the rich accumulated experience in related areas, we aim to complement existing work with new factors based on the market microstructure.

Class weighting [20], synthetic oversampling [17], and ranking losses offer different responses to class imbalance. SMOTE interpolates minority observations and can be sensitive to high-dimensional feature geometry [19]; whether that harms target prediction or score-based allocation is an empirical question. Focal loss [21] and anomaly detection [22] provide further alternatives, but are not evaluated here. Our task has an additional structure: each event cross-section contains exactly one labeled target. This motivates a within-event probability distribution rather than treating each candidate as an unrelated binary observation.

The distinction between ordering and confidence becomes operational in a portfolio. Equal investment in a deterministic top-K set depends only on rank. Score-proportional investment and weighted asset selection also depend on score separation. We therefore compare these allocation rules on the

comparison partition and motivate an event-level objective that assigns substantial probability to the true target. This is a retrospective methodological comparison, not a claim that the portfolio experiment was prospectively completed before the objective was chosen.

This paper builds upon previous work in [16] and [8], which focused on predicting manipulation targets among traded assets. We identify several critical limitations in data preparation that limit model reliability and propose innovative solutions to address these issues. Specifically, we introduce novel microstructure-based features leveraging insights from areas of market prediction [2], [13] and wash trading detection [5]. Moreover, we explore the use of boosted trees, as implemented in CatBoost [15], which has been shown to outperform other algorithms on tabular data (e.g., [7]), and we compare their performance to the random forest model introduced in [3].

Our contribution is to connect the event-level prediction objective with the portfolio rule. We distinguish deterministic rank accuracy, target-inclusion probability under weighted draws, and capital allocation after selection. We evaluate inclusion probabilities analytically without replacement and compare fixed models with paired event-level uncertainty estimates. This separates confidence-sensitive model selection from simulation noise and from the distinct question of realized portfolio returns.

Table I summarizes the key methodological differences between our framework and the most closely related prior studies.

II. DATA COLLECTION AND PREPROCESSING

A. LABELED P&DS DATA

The released event manifest, `resources/pumps.json`, contains 517 unique Binance spot BTC-pair events between January 3, 2018 and August 14, 2022. Each record supplies a target pair, exchange, and UTC event anchor. The prediction task is conditional on a scheduled event: identify its target 15 minutes before the anchor, rather than detect an unknown event time.

a: Label provenance and review.

The current release incorporates 43 additions supported by archived schedules and target announcements, one recovered historical ticker alias, and the removal of 24 duplicate or out-of-scope records. The exclusions comprise eight duplicate aliases or misspellings and 16 records outside the Binance BTC-pair scope. The evidence and exclusion manifests retain source references and review decisions. For the reviewed additions and recovered event, the anchor is the published UTC schedule known before the decision time; a subsequent price maximum is never used to choose the anchor. The remaining original labels have not undergone a comprehensive new source audit, and their retained anchors are unchanged. No minimum realized return or profitability threshold is used for admission.

b: Market-data eligibility.

For an event at T, both the labeled target and every candidate must have an actual trade in $[T - 15\text{min} - 1\text{d}, T - 15\text{min})$. We discover candidates from local historical partitions and check timestamps within that interval; partition membership or a present-day listing is insufficient. Missing-feature events are recorded separately from annotation exclusions. All 517 active events are materialized in the current experiment, producing 100,378 candidate-event observations. Restriction to one exchange and BTC quote currency supports consistent trade-level features, but limits external validity.

c: Chronological partitions.

We assign complete event cross-sections chronologically to approximately 60% training, 10% tuning, 20% comparison, and 10% test data, keeping simultaneous event groups together. Table II reports the realized partitions. Training fits model parameters; tuning supplies hyperparameter optimization and early stopping; comparison selects the model and supports the allocation analysis; test evaluates the resulting fixed choices. The comparison partition is named `VALIDATION` in exported files. Earlier research inspected these calendar periods, so this repartition is retrospective and the test is not newly unseen confirmatory evidence.

B. MARKET DATA

To enable comparison with existing research, we selected Binance as the sole exchange platform. We use Binance's public historical trade archive for every BTC-quoted pair needed by the 2018–2022 event manifest and its 30-day feature lookbacks.

A representative example of our collected data is presented in Table III.

C. FEATURE ENGINEERING

We collected trade-level data as compressed daily archives for the required Binance pairs and dates. Because trading activity varies substantially across tickers and over time, the raw event streams are irregular and cannot be used directly as fixed-size inputs to standard machine-learning models. We therefore implemented a feature-generation pipeline that (i) decompresses and loads the relevant archives, (ii) partitions data into time-aligned windows, (iii) computes microstructure features summarizing pre-announcement order flow, and (iv) aggregates ticker-level features into an event-level cross-section associated with each pump. The pipeline is:

1) Download trade-level data from Binance's public daily archive.
2) For each pump event, construct the cross-section from tickers with at least one trade in the exact interval $[T - 15\text{min} - 1\text{d}, T - 15\text{min})$. Partition membership alone is insufficient because a daily partition can contain trades later than the decision time.
3) For each eligible ticker, load 30 days of history ending at the decision cutoff and compute features over predefined half-open lookback windows $[T - 15\text{min} - x, T -$

TABLE I. Comparison of the proposed framework with closely related prior work on P&D target prediction

Study	Exchange	# Events	Model	Imbalance Handling	Backtesting
Xu & Livshits [16]	Cryptopia	180	Random forest	None	Unrealistic assumptions
Hu et al. [8]	Multiple	445	Seq. model + channel embed.	None	None
Fantazzini & Xiao [6]	Binance	Varies	Random forest	SMOTE	None
La Morgia et al. [10]	Multiple	1111	Random forest	None	None
Ours	Binance	517	Classifiers and event rankers	Cross-sectional norm., class weighting, QuerySoftMax	Equal / score-proportional allocation

TABLE II. Current chronological event partitions. Candidate rows include one positive per event.

Partition	Events	Candidate rows	First event (UTC)	Last event (UTC)
Training	310	47,468	2018-01-03 18:00:00	2020-07-10 16:00:07
Tuning	52	11,207	2020-07-10 17:00:00	2021-01-02 21:00:00
Comparison	103	26,210	2021-01-03 17:00:00	2021-08-22 17:00:00
Test	52	15,493	2021-08-24 15:00:03	2022-08-14 16:00:05

	price	qty	time	isBuyerMaker	quote
0	0.000002	2437.00	2022-07-15 15:00:02.422000	True	0.004240
1	0.000002	275.00	2022-07-15 15:00:02.434000	False	0.000479
2	0.000002	1125.00	2022-07-15 15:00:22.884000	True	0.001946
3	0.000002	941.00	2022-07-15 15:00:52.277000	True	0.001628
4	0.000002	184.00	2022-07-15 15:00:52.277000	True	0.000318

TABLE III. Structure of collected data: *price* shows the price that the order was filled at, *qty* is the amount of base asset traded, *isBuyerMaker* indicates if the buyer's order related to this transaction was already in the orderbook: if True, the trade is classified as a Sell event and if False the trade is characterized as a Buy event.

15min). Hourly normalizer bars are re-anchored to the cutoff so every included bar is strictly pre-decision.

4) Combine features for all tickers into a single event-level dataset and assign labels by adding *is_pumped*, which equals 1 for the pumped ticker and 0 otherwise.

The implementation and reproduction instructions are available in [18]. The current specification has 81 predictors: eight feature families over ten windows, plus lifetime prior-pump count. Feature generation records the actual input timestamp bounds per event, asset, feature, and dependency, including lagged prices and normalizer history. Post-announcement outcome fields are separately labeled and excluded from predictors. Small synthetic-data tests verify cutoff exclusion, train-only preprocessing, and the separation of model fitting from comparison/test data. These checks audit the implemented data flow rather than establishing that every source annotation is correct. We group features into the following categories.

1) Imbalance features

Volume imbalance measures which side is more prevalent in trade data. It is the ratio of signed notional in the quote asset (negative for seller-initiated trades and positive for buyer-initiated trades) to total absolute quote notional. Values near 1 indicate predominantly buyer-initiated flow and values near -1 predominantly seller-initiated flow.

$$\text{Volume Imbalance} = \frac{\sum_{t\in\mathcal{T}} s_t p_t q_t}{\sum_{t\in\mathcal{T}} p_t q_t}, \tag{1}$$

where $p_t q_t$ is quote notional and $s_t \in \{-1, 1\}$ is the taker-side sign at time t.

2) Quote slippages

We also created features based on slippage. Since we do not have information about order types, we cannot determine whether a trade was executed via a market or limit order; therefore, we treat slippage as a useful proxy for measuring market participants' impatience. We follow the methodology used in prior work by grouping ticks by execution time. Given that we deal with highly illiquid markets—where the time

between trades can be as long as 15 minutes—we can reasonably assume that aggregated ticks correspond to a single order. This allows us to estimate the price impact of the aggregated order. The intuition is as follows: if a market participant is willing to incur large slippage losses by submitting a large buy order, they likely want to build their position quickly, potentially in anticipation of the pump. Based on this idea, we compute slippage for each aggregated trade using the formula in (2) below:

$$L = \left| \sum_{i=1}^{N} q_i P_i - P_0 \sum_{i=1}^{N} q_i \right|, \qquad L_{\text{signed}} = L \ \text{sign}\left(\sum_{i=1}^{N} s_i q_i \right). \tag{2}$$

where q_i, P_i, and s_i are respectively the unsigned base quantity, execution price, and taker-side sign of the ith print in an aggregated timestamp group, and P_0 is its first execution price. Thus L is the absolute difference between observed quote notional and the notional implied by filling the whole group at its first price; the sign is attached from net buyer- or seller-initiated base quantity. The slippage-imbalance feature is $\sum L_{\text{signed}} / \sum L$. As an order consumes liquidity it can appear as a sequence of prints at worsening prices, so this statistic is intended to proxy how aggressively the group walks available liquidity.

3) Arithmetic return features

For each horizon x, the implementation computes the arithmetic return $10^4(p_{\text{last}}/p_{\text{prev}} - 1)$ over $[T - 15\text{min} - x, T - 15\text{min})$. Return z-scores compare that window with the mean and sample standard deviation of decision-anchored hourly returns over the preceding 30 days. For $x \in \{5, 15\}$ minutes, the exact-window return is converted to an hourly rate before standardization; for horizons of at least one hour, complete cutoff-anchored hourly bars are averaged. Thus the 5-minute, 15-minute, and 1-hour columns are distinct, while every numerator and normalizer remains strictly pre-decision.

4) Power law features. Quantile spread of order sizes

To summarize the tail of aggregated trade notionals $x \geq x_{\min}$ within each lookback window, we use the Pareto density

$$f(x; \alpha, x_{\min}) = (\alpha - 1) x_{\min}^{\alpha - 1} x^{-\alpha}, \qquad \alpha > 1, \tag{3}$$

where $x_{\min}$ is the minimum positive aggregated notional observed in the window. The code uses the closed-form maximum-likelihood estimate

$$\hat{\alpha} = 1 + \frac{n}{\sum_{i=1}^{n} \log(x_i / x_{\min})}. \tag{4}$$

Smaller $\hat{\alpha}$ indicates a heavier notional tail. Before cross-sectional standardization, the implementation clips $\hat{\alpha}$ to the closed interval $[1, 2]$. This deterministic bound handles degenerate short windows in which the denominator is zero (and the raw estimate is infinite) and limits the influence of extremely sparse tails. This is a fixed model-preprocessing rule, not a sample-dependent 95% winsorization; no histogram-density regression or log–log OLS is performed.

All of the features discussed are further explained below (see Table IV):

D. CROSS-SECTIONAL APPROACH

Given that our event dataset spans early 2018 through August 2022, we account for temporal heterogeneity by applying cross-sectional standardization. This approach reduces market-regime level shifts and focuses the models on relative cross-asset dynamics.

We apply cross-sectional standardization to the unbounded return, return-z-score, quote-volume-z-score, and power-law features. For each selected feature, we subtract its mean within an event and divide by its within-event sample standard deviation:

$$X_{\text{standardized}} = \frac{X - \bar{X}_{CS}}{\bar{\sigma}_{CS}} \tag{5}$$

If a selected feature is constant within an event, it contains no relative ranking information and is mapped to zero; all-missing event columns remain missing for native handling or train-only imputation. We do not apply this second-stage transform to bounded imbalance/share features or to count features (trade counts and lifetime prior-pump count), whose nonnegative levels retain their direct interpretation. Fig. 2 demonstrates the transform on a sample of five cross-sections for three selected numerical features:

III. CONFIDENCE-SENSITIVE MODELLING

For event e, let N_e be the number of eligible candidates and y_e the single labeled target. All models use the same pre-decision predictors. The eight fitted families are logistic regression, random forest, a class-weighted CatBoost classifier, CatBoost with SMOTE, a return-target CatBoost ranker, a CatBoost classifier with rank-AUC early stopping, CatBoost with QuerySoftMax, and a custom Top-5 margin objective. Each family uses its saved tuning-only hyperparameter search; the comparison is between one tuned representative per family.

The return-target ranker uses post-announcement five-minute returns as training labels, ordered within each event. QuerySoftMax instead uses the binary target identity, with one positive per event. Future returns are labels for the former model and outcomes for portfolio evaluation, never predictor columns. Accordingly, a difference between these two rankers cannot be attributed solely to the ranking algorithm: their supervision differs.

A. WHY THE LOGIT GAP MATTERS

For QuerySoftMax scores z_{ej}, define a within-event distribution

$$p_{ej} = \frac{\exp(z_{ej})}{\sum_{\ell=1}^{N_e} \exp(z_{e\ell})}. \tag{6}$$

Feature	Description	Notation
Asset return	return of the asset calculated over the window of size (T-15min-x, T-15min) prior to the pump announcement	*asset_return*[x]@t
Standardized return	exact-window hourly-rate return (5m, 15m) or mean cutoff-anchored hourly return ($x \geq$ 1h), minus the 30-day hourly mean and divided by its standard deviation	*asset_return_zscore*[x]@t
Standardized volume in BTC	exact-window hourly-rate volume (5m, 15m) or mean cutoff-anchored hourly volume ($x \geq$ 1h), standardized by 30-day hourly moments	*quote_abs_zscore*[x]@h
Share of long trades	share of long trades as the ratio of number of long trades and all trades of this time window (T-15min-x, T-15min)	*share_of_long_trades*[x]@h
Number of trades	number of timestamp-aggregated trades within $[T - 15\text{min} - x, T - 15\text{min})$	*num_trades*[x]@h
Flow imbalance ratio	volume imbalance ratio computed over the window of size (T-15min-x, T-15min)	*flow_imbalance*[x]@h
Quote slippage imbalance ratio	imbalance ratio of slippages in BTC calculated over the same time window	*slippage_imbalance*[x]@t
Powerlaw alpha	estimated alpha from powerlaw probability density function using the window of size (T-15min-x, T-15min) prior to the pump	*powerlaw_alpha*[x]@h
Number of previous pumps	lifetime number of manifest events for this token strictly before the decision cutoff $T - 15\text{min}$	*num_prev_pump*

TABLE IV. These are the features included in all models. These features are used to describe historical dynamics of price and trading volume in the run up to the pump. $x \in \{5m, 15m, 1h, 2h, 4h, 12h, 1d, 2d, 7d, 14d\}$

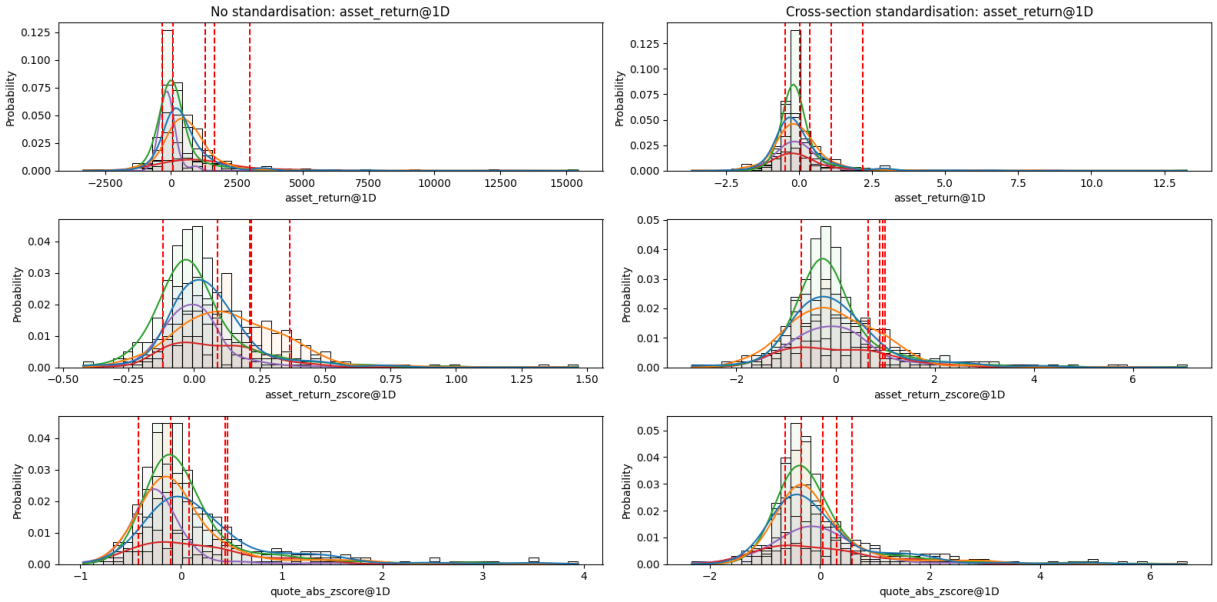


FIGURE 2. Features with and without cross-sectional standardization: histograms and kernel-density estimates for five P&D events. Applying standardization helps to align distributions more closely with one another across pumps.

A model can rank the true target first with only a tiny score advantage. For example, if its logit exceeds $N_e - 1$ tied alternatives by δ, then $p_{e,y_e} = e^{\delta}/(e^{\delta} + N_e - 1)$: correct Top@1 rank alone does not imply a large target probability. In a weighted portfolio, that distinction affects how often the target is selected and how much capital it receives.

With one positive label per event and temperature parameter fixed at one, the event-level QuerySoftMax objective is

$$\mathcal{L} = -\frac{1}{E}\sum_{e=1}^{E} \log p_{e,y_e} \tag{7}$$
$$= \frac{1}{E}\sum_{e=1}^{E} \log\left(1 + \sum_{j \neq y_e} \exp(z_{ej} - z_{e,y_e})\right).$$

Thus the loss continues to reward separation after the target reaches first place. It increases the target score relative to all alternatives, rather than rewarding confidence in an arbitrary predicted winner. It does not guarantee calibrated probabilities, larger gaps on unseen events, or higher financial returns. Here, "confidence" denotes the concentration of the model-implied within-event distribution.

B. TRAINING AND SELECTION ROLES

All model fits use only the training partition. Optuna searches [1] and early stopping use only tuning events. QuerySoftMax minimizes mean event log loss on tuning, with 50-round early stopping; it does not directly optimize the analytical inclusion AUC used later for model selection. The other families retain their own tuning criteria, including rank-based metrics for the relevant classifiers and the Top-5 objective. The saved experiments use 100-trial searches per family. Imputation and fitted preprocessing statistics are learned from training data, while cross-sectional standardization uses only the contemporaneous pre-decision candidate features of each event.

Once fitted, models are held fixed. The highest analytical inclusion AUC on the comparison partition selects the reported model. Test outcomes do not change that selection in the current run. Because both the research design and the historical partitions have been inspected during earlier work, the analysis should be interpreted as a reproducible retrospective evaluation rather than a prospectively preregistered experiment.

IV. PORTFOLIO CONSTRUCTION AND EVALUATION

A. ASSET SELECTION AND CAPITAL ALLOCATION

We distinguish selecting a set of K assets from allocating capital within that set. Let $w_{ej} \geq 0$ denote a model's selection weight. For the binary classifiers, w_{ej} is the positive-class predicted probability. For the ranking and margin models, $w_{ej} = \exp(z_{ej})$, computed with a stable softmax normalization when needed. Raw logits themselves are not monetary weights.

We evaluate both the deterministic K highest-scoring assets and successive weighted draws without replacement. In the latter case,

$$\Pr(J_r = j \mid J_1, \ldots, J_{r-1}) = \frac{w_{ej}}{\sum_{\ell \notin \{J_1,\ldots,J_{r-1}\}} w_{e\ell}}. \tag{8}$$

Each selected asset therefore appears once. Zero-weight candidates are considered only after all positive-weight candidates have been selected; any remaining slots are filled uniformly. An all-zero event uses uniform sampling. Given the selected set S_{eK}, the paired allocation comparison uses

$$a_{ej}^{\text{equal}} = \frac{1}{K}, \qquad a_{ej}^{\text{score}} = \frac{w_{ej}}{\sum_{\ell \in S_{eK}} w_{e\ell}}, \quad j \in S_{eK}. \tag{9}$$

An all-zero selected set receives equal capital weights. The same assets and execution outcomes are used for both allocations. This isolates capital weighting from selection. Changing allocation alone cannot change the probability that the set contains the target. For QuerySoftMax, score-proportional allocation is softmax allocation restricted to the selected set; "logit-weighted" below refers to this exponential transformation, not direct normalization of possibly negative logits.

B. EXECUTION AND PORTFOLIO SIMULATIONS

Entries are submitted at $T - 15$ minutes and filled at the first subsequent trade strictly before T. A selected non-target exits at the first trade at or after T; the announced target exits at the first trade at or after $T + 1$ minute. This exit rule assumes that target identity is available at T; target-reveal delays are not modeled. The identity is never used for pre-announcement selection. Proceeds are converted from BTC to USDT using the implemented indicative rates, and filled legs incur a 25-basis-point round-trip cost. A leg without valid execution retains its assigned capital as zero-return cash; neither replacement sampling nor redistribution occurs.

For portfolio returns and drawdowns, we simulate 1,000 complete event-sequence paths for each model and $K = 1, \ldots, 30$. Shared Gumbel perturbations implement Eq. 8, pairing models and allocations on identical random draws; nested prefixes pair different values of K. Each event has the same fixed notional budget, with no reinvestment. Reported cumulative return is the sum of event returns, and maximum drawdown is measured on $1 + \sum r_e$ at event boundaries, combining simultaneous events before marking equity. The return and drawdown summaries are separate medians across complete paths.

These simulations quantify selection randomness conditional on fixed events, prices, and fitted models. Their percentiles are not sampling confidence intervals or hypothesis-test p-values. The current allocation comparisons charge the stated transaction cost but do not include fitted price impact, latency, or order-book execution. Earlier impact experiments using a different model and split are not transferred to the current results.

C. ANALYTICAL TARGET-INCLUSION PROBABILITIES

For model m, define $H_{em}(K) = \Pr(y_e \in S_{eK})$. With no replacement and one true target, this is the probability of selecting that target at least once in K draws. It depends on the entire cross-section of weights, not only the target's initial probability. The independent-draw expression $1-(1-p_{e,y_e})^K$ is inapplicable.

For a positive-weight target, an exponential-race representation gives a deterministic calculation. Draw independent conceptual clocks $\tau_j \sim \text{Exp}(w_{ej})$ and take the K smallest. Conditional on the target clock being t, each competing clock precedes it independently with probability $1 - e^{-w_{ej}t}$. Therefore,

$$H_e(K) = \int_0^\infty w_{ey} e^{-w_{ey}t} F_e(t, K-1)\, dt, \tag{10}$$

$$F_e(t, K-1) = \sum_{r=0}^{K-1} [u^r] \prod_{j \neq y} \left[e^{-w_{ej}t} + (1 - e^{-w_{ej}t})u\right].$$

Here $[u^r]$ extracts a polynomial coefficient. If the target has zero weight and M candidates have positive weight, its inclusion probability is $\max(0, K-M)/(N_e-M)$ for $K \le N_e$; an all-zero event gives K/N_e. The probability law is exact; the implementation evaluates the integral numerically using log-time quadrature and a truncated Poisson-binomial recurrence, with tolerance 2×10^{-10}. No sampled portfolios enter this calculation. For positive total weight, $H_e(1) = w_{ey}/\sum_j w_{ej}$; when all candidates are selected, $H_e(N_e) = 1$.

We average $H_e(K)$ across events. To account for changing cross-section sizes, the percentage curve uses $K_e(f) = \lceil fN_e \rceil$ for $f > 0$ and $K_e(0) = 0$. The primary selection metric is

$$\mathrm{Top@K\%AUC_{incl}} = \frac{1}{0.20}\int_0^{0.20} \frac{1}{E}\sum_e H_e(K_e(f))\,df. \tag{11}$$

We use trapezoidal integration on $f = 0, 0.005, \ldots, 0.20$. The low-percentage range emphasizes concentrated portfolios. This analytical inclusion AUC is distinct from deterministic Top@K rank accuracy: at $K = 1$ it averages assigned target probability, not the indicator that the target ranks first. The latter remains a useful descriptive diagnostic but does not measure probability concentration.

D. PAIRED STATISTICAL INFERENCE

We resample whole pump events with replacement, keeping every candidate within its event and using the same resampled event indices for all eight fixed models. For each partition, 10,000 bootstrap replicates (seed 42) yield pointwise percentile 95% intervals for mean event inclusion AUC and paired differences. Two-sided tests use centered bootstrap difference distributions with a finite-replicate correction. Holm adjustment covers all $\binom{8}{2} = 28$ model pairs separately on comparison and test; displaying only QuerySoftMax comparisons does not reduce the correction family.

The bootstrap concerns variation across events, while the probability integral removes portfolio-sampling noise from each event's inclusion score. It excludes retraining, tuning, model-selection uncertainty, and serial dependence between events. Comparison-partition inference is additionally conditional on the model-selection exercise. Statistical claims below refer to inclusion AUC, not to portfolio profit or universal superiority under other objectives.

V. RESULTS

A. ALLOCATION EVIDENCE AND CONFIDENCE-SENSITIVE PREDICTION

On the 103-event comparison sample, QuerySoftMax's sampled $K = 5$ portfolio has median cumulative return 88.15% under equal allocation and 213.07% under score-proportional allocation. Median maximum drawdown changes from 7.22% to 7.34%. Holding its deterministic top-five set fixed gives cumulative returns of 182.94% and 224.90%, respectively. Thus the comparison demonstrates that assigning more capital to higher scores can materially change the result even with the same selected assets; it also shows why correct rank alone is an incomplete training target for this rule.

On the 52-event test sample, QuerySoftMax's sampled median cumulative return rises from 23.12% to 81.58% under score-proportional allocation, while median drawdown rises from 5.79% to 7.65%. For its deterministic top-five set, cumulative return rises from 44.96% to 89.29%, with drawdown increasing from 3.39% to 4.38%. The test therefore reproduces the direction of the return improvement but also exposes a concentration-risk trade-off. These are conditional portfolio outcomes, not statistical tests of the allocation effect.

QuerySoftMax has the highest score-proportionally allocated sampled $K = 5$ median return among the eight models on both partitions (Fig. 3). This statement is specific to that selection and allocation rule. On test, SMOTE's corresponding return is 78.59% with lower median drawdown, 5.76%; QuerySoftMax does not dominate it on both return and risk. Under deterministic top-five selection, the standard CatBoost classifier instead has the highest score-proportional test return, 102.24%. These differences motivate explicit reporting of the portfolio rule rather than declaring one model best for every portfolio.

B. COMPARISON SELECTION AND TEST INCLUSION PERFORMANCE

QuerySoftMax has the highest comparison analytical inclusion AUC, 0.4301 (95% CI [0.3731, 0.4879]), narrowly ahead of SMOTE at 0.4252. It is therefore the selected model. Its test inclusion AUC is 0.5036 (95% CI [0.4250, 0.5819]). SMOTE has the highest test point estimate, 0.5555, but the test ranking does not change the comparison-selected model. The other six models have lower AUC on both partitions (Table VI).

The objective distinction is substantive. QuerySoftMax optimizes probability assigned to the labeled target. The return-target ranker and the Top-5 margin model optimize different properties, while the rank-AUC early-stopping classifier can preserve ordering without producing a concentrated sampling distribution. Their lower inclusion AUC does not establish that their deterministic ranks or all other uses of their scores are inferior. Similarly, the current evidence does not support a general claim that SMOTE fails: it is the strongest alternative on this confidence-sensitive metric.

C. WHICH DIFFERENCES ARE STATISTICALLY SUPPORTED?

For QuerySoftMax versus each of logistic regression, random forest, the standard CatBoost classifier, the return-target ranker, the rank-AUC early-stopping classifier, and the Top-5 model, the analytical inclusion-AUC difference is positive on both partitions, with Holm-adjusted $p = 0.0028$. For example, the advantage over logistic regression is 0.0953 on comparison (paired 95% CI [0.0530, 0.1383]) and 0.1382 on test (paired 95% CI [0.0744, 0.2036]). These paired tests establish statistically supported improvements over six evaluated alternatives under the stated event-bootstrap assumptions.

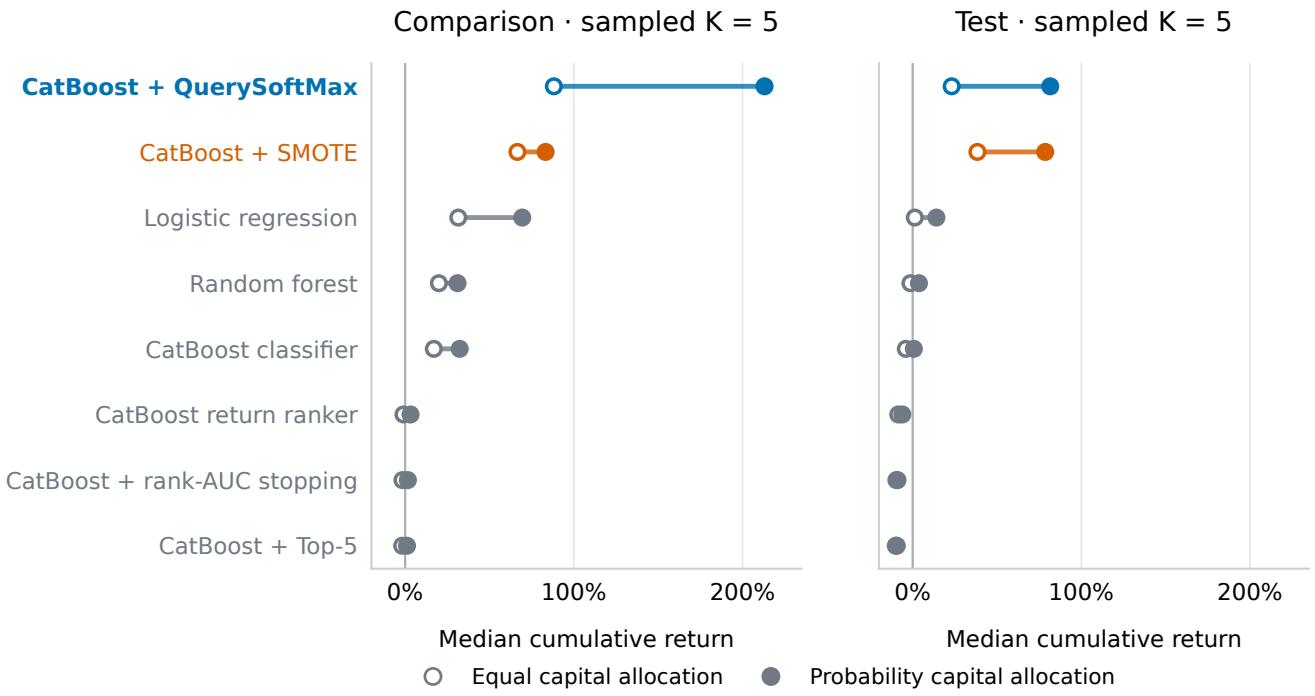


FIGURE 3. Equal and score-proportional capital allocation for the same sampled five-asset sets, by model and partition. Points show median fixed-notional cumulative return across 1,000 complete paths. Connecting lines isolate the allocation change; they are not confidence intervals. Comparison and test span different numbers of events and calendar periods.

TABLE V. Sampled $K = 5$ portfolios: equal → probability capital allocation. Returns and maximum drawdowns are separate medians across 1,000 complete paths, in percent. Returns sum fixed-notional event returns without compounding. These are conditional Monte Carlo summaries, not statistical confidence intervals.

	Comparison: 103 events		Test: 52 events	
Model	Return	Max. drawdown	Return	Max. drawdown
CatBoost + QuerySoftMax	88.15 → 213.07	7.22 → 7.34	23.12 → 81.58	5.79 → 7.65
CatBoost + SMOTE	66.53 → 83.26	8.12 → 10.02	38.38 → 78.59	4.75 → 5.76
Logistic regression	31.48 → 69.42	12.98 → 19.44	1.28 → 14.08	10.32 → 13.92
Random forest	19.96 → 31.04	11.39 → 13.08	−1.34 → 3.72	9.34 → 9.61
CatBoost classifier	16.98 → 32.31	11.55 → 13.63	−4.06 → 0.69	10.34 → 10.58
CatBoost return ranker	−1.02 → 3.17	14.29 → 17.14	−8.54 → −6.14	11.73 → 12.59
CatBoost + rank-AUC stopping	−1.61 → 1.60	13.54 → 14.20	−9.66 → −8.99	11.70 → 11.89
CatBoost + Top-5	−1.78 → 1.09	13.43 → 14.70	−10.03 → −9.42	11.90 → 12.18
Uniform selection	−5.70 → −5.70	13.61 → 13.61	−10.82 → −10.82	11.99 → 11.99

TABLE VI. Analytical Top@k%AUC for weighted draws without replacement. Intervals are pointwise 95% event-bootstrap intervals (10,000 resamples). QuerySoftMax is selected on comparison performance; test ordering does not change selection.

	Comparison: 103 events		Test: 52 events	
Model	AUC	95% CI	AUC	95% CI
CatBoost + QuerySoftMax	0.4301	[0.3731, 0.4879]	0.5036	[0.4250, 0.5819]
CatBoost + SMOTE	0.4252	[0.3576, 0.4930]	0.5555	[0.4595, 0.6468]
Logistic regression	0.3348	[0.2970, 0.3739]	0.3654	[0.3257, 0.4052]
Random forest	0.2905	[0.2635, 0.3175]	0.3039	[0.2751, 0.3300]
CatBoost classifier	0.2537	[0.2357, 0.2713]	0.2546	[0.2338, 0.2733]
CatBoost return ranker	0.1422	[0.1331, 0.1517]	0.1584	[0.1439, 0.1733]
CatBoost + rank-AUC stopping	0.1352	[0.1312, 0.1391]	0.1383	[0.1334, 0.1428]
CatBoost + Top-5	0.1254	[0.1156, 0.1362]	0.1274	[0.1131, 0.1441]

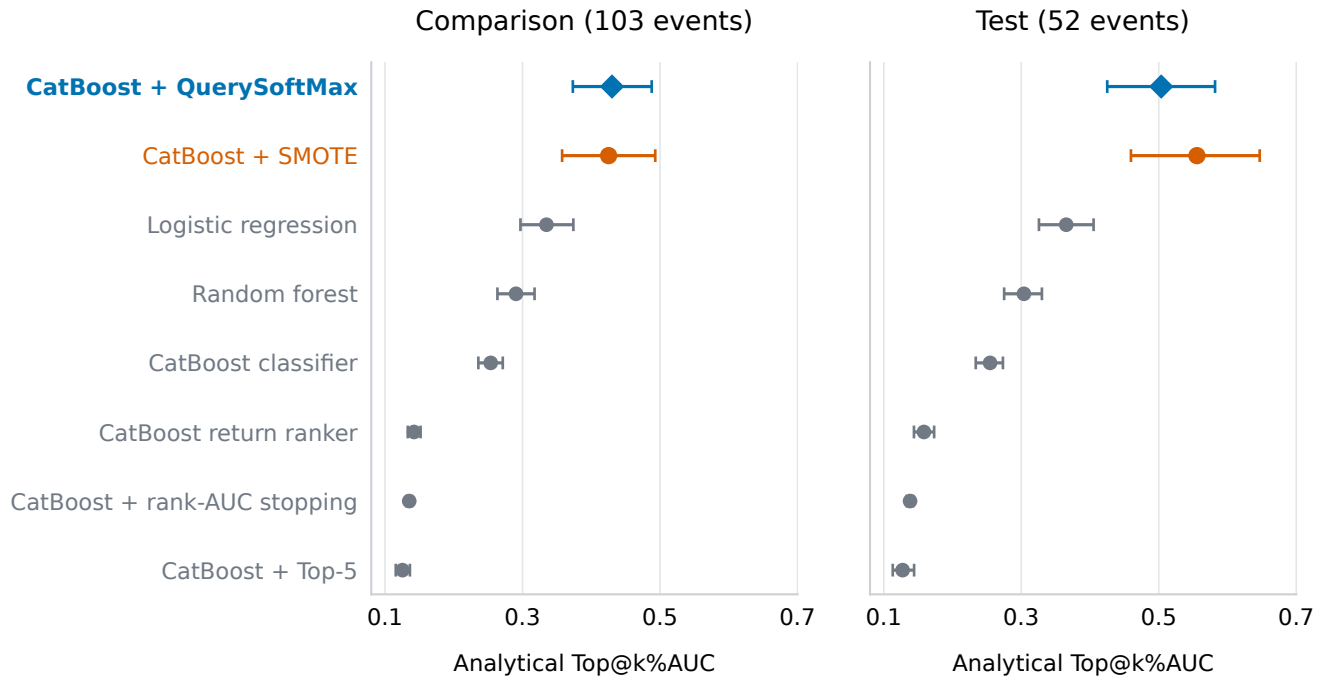


FIGURE 4. Analytical target-inclusion AUC under weighted draws without replacement, with pointwise 95% whole-event bootstrap intervals. QuerySoftMax is selected on comparison; SMOTE has the higher test point estimate. Model-difference significance is assessed by paired tests with Holm correction, not by overlap of these marginal intervals.

TABLE VII. QuerySoftMax minus each comparator: analytical Top@k%AUC differences, pointwise 95% paired bootstrap intervals, and two-sided *p*-values adjusted by Holm over all 28 model pairs separately within each partition. Positive differences favor QuerySoftMax.

	Comparison: 103 events			Test: 52 events		
Comparator	Difference	95% CI	p_{Holm}	Difference	95% CI	p_{Holm}
CatBoost + SMOTE	$+0.0049$	$[-0.0524, +0.0623]$	0.8657	-0.0519	$[-0.1271, +0.0188]$	0.2380
Logistic regression	$+0.0953$	$[+0.0530, +0.1383]$	0.0028	$+0.1382$	$[+0.0744, +0.2036]$	0.0028
Random forest	$+0.1397$	$[+0.0979, +0.1822]$	0.0028	$+0.1997$	$[+0.1329, +0.2677]$	0.0028
CatBoost classifier	$+0.1764$	$[+0.1310, +0.2225]$	0.0028	$+0.2489$	$[+0.1826, +0.3159]$	0.0028
CatBoost return ranker	$+0.2880$	$[+0.2362, +0.3400]$	0.0028	$+0.3452$	$[+0.2707, +0.4188]$	0.0028
CatBoost + rank-AUC stopping	$+0.2949$	$[+0.2410, +0.3498]$	0.0028	$+0.3653$	$[+0.2899, +0.4404]$	0.0028
CatBoost + Top-5	$+0.3047$	$[+0.2505, +0.3590]$	0.0028	$+0.3762$	$[+0.3025, +0.4496]$	0.0028

The nearest competitor is an important exception. QuerySoftMax minus SMOTE is 0.0049 on comparison (paired 95% CI $[-0.0524, 0.0623]$, adjusted $p = 0.8657$) and -0.0519 on test (paired 95% CI $[-0.1271, 0.0188]$, adjusted $p = 0.2380$). Neither difference is significant at 5%. Consequently, the evidence supports QuerySoftMax as the comparison-selected model with significant gains over six alternatives, not as statistically superior to all alternatives or the highest-AUC model on test.

D. REPRODUCIBILITY AND INTERPRETATION

The reported inclusion results come from the current without-replacement analytical export, recomputed from saved frozen-model predictions; earlier Monte Carlo and with-replacement inclusion exports are superseded. The current predicted event scores reproduce the saved analytical AUCs and paired bootstrap results to numerical precision. The allocation figures use the paired equal-versus-probability exports for the same fitted models. Source paths and file hashes accompanying the paper figures identify these artifacts. This distinction prevents mixing earlier model fits, splits, or probability laws in one results table.

Taken together, the allocation comparison and inclusion analysis support the central mechanism: once scores determine both selection and capital concentration, model quality depends on how much probability the true target receives, beyond its rank. QuerySoftMax directly addresses that requirement. The empirical result remains criterion-dependent and retrospective, and its portfolio return advantage requires separate statistical and execution validation.

VI. LIMITATIONS AND FUTURE WORK

a: Sample size and historical reuse.

There are 517 events in total and only 52 in the test partition. Earlier research inspected these periods, and the current analysis combines objective design, allocation exploration, and

model comparison. A genuinely new chronological sample is needed for confirmatory validation. Whole-event bootstrapping preserves within-event dependence but assumes independent events; clustered events, repeated assets, and shared market regimes may reduce effective sample size. The fixed-model intervals omit the uncertainty of retraining and selecting a model.

b: Confidence and calibration.

A concentrated softmax distribution is not proof of calibrated manipulation probabilities. Score scale affects both inclusion and allocation, and excessive confidence in a wrong target can be costly. Temperature calibration fitted only on tuning data, explicit concentration controls, and alternative proper scoring objectives are natural extensions. The present results do not demonstrate that simply increasing every model's logit gaps would improve performance.

c: Execution and risk.

The allocation experiment includes a fixed transaction-cost allowance but no market-impact model. Event-boundary drawdown omits intratrade losses, and fixed per-event budgets do not model capital constraints when positions overlap. The availability and latency of the public target reveal also affect exit feasibility. Consequently, the reported cumulative returns are backtest outcomes under specified assumptions, not evidence of deployable profitability or capacity.

d: Labels, features, and external validity.

Only the reviewed additions and recovered ticker have the documented new source audit; the remaining labels retain their historical anchors. Some public source exports omit edit histories. Timestamp provenance and dummy-data tests help verify that predictor calculations precede the decision cutoff, but cannot eliminate annotation error or all forms of research-design bias. Binance BTC-pair results need validation on other venues, quote currencies, and periods. Alternative losses and retraining stability remain future work; robustness analyses from older fits are not evidence for the present QuerySoftMax model.

VII. CONCLUSION

Comparing equal and score-proportional portfolios shows why a model should assign substantial probability to the true manipulated asset rather than merely rank it first. Event-level QuerySoftMax training expresses that requirement through a loss on the true target's within-event softmax probability. Deterministic integration under weighted sampling without replacement then evaluates the corresponding inclusion probabilities without simulating portfolios.

QuerySoftMax achieves the highest comparison analytical inclusion AUC and significantly exceeds six alternatives on both comparison and test after multiple-comparison correction. Its sampled, score-proportionally allocated five-asset portfolios also have the highest median return among the eight evaluated models on both partitions. However, SMOTE has the higher test inclusion AUC and is not significantly distinguishable from QuerySoftMax, and portfolio return differences are not established as statistically significant. These results favor confidence-sensitive learning for the examined portfolio rule while leaving calibration, prospective validation, execution capacity, and uncertainty in the allocation benefit open for further study.

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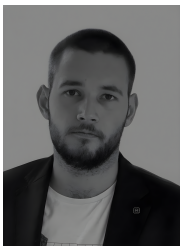
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